Ranked Set Sampling under Imperfect Ranking: Design-Based Analysis and Adaptive Selection

Jaafar Almutawa *
Department of Mathematics, Bahrain Polytechnic
email: jaafar111@yahoo.com

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Abstract

Ranked Set Sampling (RSS) can improve precision when observations are expensive to measure but inexpensive to rank, but imperfect ranking can change both the sampling distribution and the resulting efficiency. We formulate imperfect ranking through a rank-transition matrix rather than treating a scalar rank correlation as a complete description of ranking quality. Under an explicit balanced-RSS model and a stated doubly stochastic condition on the transition matrix, we derive exact finite-sample expressions for the distribution and variance of the RSS mean and identify the corresponding perfect-ranking and SRS limits. We then discuss robust estimation after RSS selection, emphasizing that robustness of a base estimator does not by itself imply robustness of the combined selection-and-estimation procedure. Finally, we introduce an adaptive RSS–SRS rule that uses an independent pilot to estimate variance and can reject RSS when its estimated precision is insufficient. A final V3 simulation study evaluates this rule over 96 univariate conditions formed by four ranking-quality targets, two ranking-error mechanisms, two contamination levels, two contamination-placement mechanisms, and three estimators. With a 200-replication pilot, 1,000 adaptive evaluation trials per condition, and an independent 5,000-replication fixed-strategy reference experiment, RSS was selected in 83.97% of trials and agreed with the ex-post variance oracle in 83.54% of trials. The results support a conditional, mechanism-dependent interpretation of RSS efficiency rather than a universal dominance claim.

Keywords: High-Breakdown estimators, Ranked Set Sampling, Statistical Efficiency, Contaminated Data Models

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1 Introduction

Robust estimation is designed for settings in which a non-negligible fraction of observations may be atypical, corrupted, or otherwise inconsistent with the assumed model. High-breakdown procedures such as the Minimum Covariance Determinant (MCD), S-estimators, Minimum Volume Ellipsoid (MVE), and Stahel–Donoho procedures provide strong protection against contamination, but this protection can come with a loss of statistical efficiency and, for some procedures, substantial computational cost **??**.

Ranked Set Sampling (RSS) provides a different way to use information. When a cheap auxiliary assessment can rank units reasonably well before the expensive response is measured, RSS deliberately selects units from different rank strata rather than measuring a simple random sample. Under perfect ranking, balanced RSS can reduce the variance of the sample mean relative to simple random sampling (SRS). The classical RSS literature also shows, however, that imperfect ranking changes the sampling distribution and that its effect depends on the ranking mechanism and the target of inference **???**.

This observation is central to the present paper. A scalar rank-correlation coefficient is useful as a diagnostic, but it is generally not sufficient to determine the sampling distribution induced by imperfect ranking. Two ranking mechanisms can have similar overall rank correlation while producing different probabilities of confusing adjacent ranks, reversing extreme ranks, or concentrating errors in particular strata. These mechanisms can therefore lead to different efficiency even when their scalar correlation is the same. We consequently formulate imperfect ranking through a rank-transition mechanism and use rank correlation only as a summary diagnostic.

The robust-estimation problem adds a second layer. Ranking is itself a selection mechanism, so a ranking rule that is strongly affected by contamination can preferentially select contaminated observations and thereby alter the effective robustness of the final estimator. Conversely, a ranking rule that is stable under contamination may preserve the practical robustness of the base estimator while still providing useful efficiency gains. The appropriate question is therefore not whether RSS is universally better than SRS, but *when the available ranking information is sufficiently informative and robust for RSS to be preferable.*

The paper develops this viewpoint in four steps. First, we specify an explicit imperfect-ranking model and derive design-based expressions for balanced RSS. Second, we connect the sampling design to robust estimators through influence-function and contamination arguments, distinguishing general consequences of the sampling design from estimator-specific guarantees. Third, we introduce an adaptive decision rule that can fall back to SRS when the estimated ranking information does not justify RSS. Fourth, we use a calibrated univariate simulation study to evaluate the resulting gain and failure regimes across ranking mechanisms, contamination placement, ranking quality, and robust estimators.

The main contributions are therefore:

1. an explicit transition-matrix formulation of imperfect ranking that replaces the unsupported assumption that a single scalar correlation determines RSS efficiency;
2. exact finite-sample mean and variance expressions for balanced RSS under stated ranking assumptions, together with clear conditions for the SRS and perfect-ranking

limits;

3. a conditional framework for combining RSS with robust estimators, with robustness claims stated only under explicit assumptions on the ranking mechanism and the estimator;
4. an adaptive RSS–SRS selection principle that recognizes failure regimes rather than assuming that RSS must improve precision; and
5. a simulation protocol designed to separate the effects of ranking mechanism, ranking quality, contamination, dimensionality, and estimator choice.

The paper is organized as follows. Section 2 reviews the robust estimators considered in the study. Section 3 develops the imperfect-ranking model and its consequences for RSS efficiency and robustness. Section 4 describes the adaptive implementation and experimental protocol. Section 5 reports the numerical evidence. Section 6 summarizes the conclusions and limitations.

2 High-Breakdown Estimators: Theory and Computational Considerations

High-breakdown estimators are indispensable tools in multivariate statistical analysis, particularly when data may be contaminated with outliers. These estimators balance robustness with statistical efficiency and are capable of handling large contamination levels without severe degradation in performance **?**. This section presents the theoretical foundations and computational properties of several widely used high-breakdown estimators for location and scatter, providing context for the later RSS experiments. Any computational benefit from RSS is treated as a separate empirical question rather than assumed from the reduction in measured observations.

Let T_n be an estimator based on a sample $X_n = \{x_1, \ldots, x_n\}$. The finite-sample breakdown point of T_n at X_n is defined as **?**:

$$\epsilon_n^*(T_n, X_n) = \frac{1}{n} \min \left\{ m : \sup_{X_n'} \|T_n(X_n') - T_n(X_n)\| = \infty \right\},$$

where X_n' is obtained by replacing m observations in X_n with arbitrary values. In other words, the breakdown point quantifies the smallest proportion of contaminated data that can cause the estimator to yield arbitrarily large errors. The asymptotic breakdown point is the limit of ϵ_n^* as $n \to \infty$, representing the estimator's ultimate robustness against increasing sample size.

For affine equivariant estimators, the following theoretical upper bounds apply **?**:

$$\epsilon_n^* \leq \frac{1}{n} \left\lfloor \frac{n+1}{2} \right\rfloor \quad \text{(location)}, \quad \epsilon_n^* \leq \frac{1}{n} \left\lfloor \frac{n-p+1}{2} \right\rfloor \quad \text{(scatter)},$$

where p is the dimension of the data. These bounds establish the fundamental limits of robustness achievable for affine equivariant estimators. The maximum attainable breakdown point of 50% is considered optimal, as it corresponds to the highest proportion of contamination that can be tolerated without complete loss of estimator reliability **?**.

High-breakdown estimators (with $\epsilon^* \approx 0.5$) generally possess the following features **?**:

- **Affine Equivariance:** For any affine transformation ($A\mathbf{x} + \mathbf{b}$), these estimators ensure consistent results across transformations, formally expressed as:

$$\hat{\boldsymbol{\mu}}(A\mathbf{X} + \mathbf{b}) = A\hat{\boldsymbol{\mu}}(\mathbf{X}) + \mathbf{b}, \quad \hat{\boldsymbol{\Sigma}}(A\mathbf{X} + \mathbf{b}) = A\hat{\boldsymbol{\Sigma}}(\mathbf{X})A^\top.$$

- **Subset or Projection-Based Resistance:** They rely on robust subsets (e.g., MCD, MVE) or projection-based diagnostics (e.g., Stahel-Donoho) to identify and downweight or exclude outliers.
- **Distribution-Free Robustness:** These estimators do not require specific distributional assumptions on the outlier-generating mechanism for their breakdown point guarantees. However, efficiency results (such as asymptotic relative efficiency) may depend on the underlying distribution and the ranking error model; see Section 3 for a detailed discussion.
- **Efficiency-Robustness Trade-off:** While achieving high breakdown points, these estimators typically exhibit reduced statistical efficiency under clean data conditions compared to classical maximum likelihood estimators. Efficiency is often measured as asymptotic relative efficiency (ARE) under the multivariate normal model.

Robust multivariate estimation is essential in practice since classical estimators are sensitive to contamination. High-breakdown estimators mitigate this issue by tolerating a significant portion of arbitrary contamination without catastrophic performance degradation. However, their reduced efficiency under ideal conditions and computational demands motivate careful consideration of the sampling design used before estimation. To provide context for the later experiments, we review four foundational high-breakdown estimators in terms of robustness, efficiency, and computational considerations.

2.1 Minimum Covariance Determinant (MCD)

The MCD estimator **?** identifies a subset $H \subset \{1, \ldots, n\}$ of size $h \approx n/2$ that minimizes the determinant of the sample covariance matrix:

$$\det(\hat{\boldsymbol{\Sigma}}_H) = \det\left(\frac{1}{h-1}\sum_{i\in H}(\mathbf{x}_i - \bar{\mathbf{x}}_H)(\mathbf{x}_i - \bar{\mathbf{x}}_H)^\top\right),$$

where $\bar{\mathbf{x}}_H = \frac{1}{h}\sum_{i\in H}\mathbf{x}_i$. The mean and covariance of this optimal subset serve as the raw MCD estimates. The MCD estimator is widely used for robust estimation of multivariate location and scatter.

A reweighting step based on Mahalanobis distances from the raw MCD estimates is typically used to improve efficiency. The FAST-MCD algorithm **?** accelerates computation, reducing complexity from $O(n^3)$ to $O(n^2 p)$ via intelligent subsampling and iterative refinement.

The breakdown point for $h = \left\lfloor \frac{n+p+1}{2} \right\rfloor$ is

$$\epsilon_n^* = \frac{n-h+1}{n} \approx \frac{n-p+1}{2n} \to 0.5 \quad \text{as } n \to \infty.$$

Thus, MCD can tolerate nearly 50% contamination. Its asymptotic efficiency at the normal model is about 85% after reweighting **?**. However, MCD requires $n > 2p$ for the estimator to be well-defined, and its efficiency is lower than that of the sample mean/covariance under clean data.

In practice, the MCD can quickly become too demanding to compute when the number of variables is large compared to the sample size, even with faster approximations such as FAST-MCD. Its performance also depends on how the subset size h is chosen and on the stability of the reweighting step, which can make results somewhat sensitive. Ranked Set Sampling changes which observations are measured, but the present study does not establish a reduction in the computational cost of these robust estimators. Any computational advantage would require a separate benchmark that accounts for ranking, sorting, estimator fitting, and precision-matched sampling cost.

2.2 S-Estimators

S-estimators **?** are defined by minimizing a robust scale σ subject to a constraint:

$$\min_{\boldsymbol{\mu},\boldsymbol{\Sigma}} \sigma \quad \text{subject to} \quad \frac{1}{n}\sum_{i=1}^{n} \rho\left(\frac{d_i^2}{\sigma^2}\right) = \delta,$$

where d_i is the Mahalanobis distance, ρ is a bounded loss function (e.g., Tukey's biweight), and δ is a constant for consistency. S-estimators achieve breakdown points approaching 0.5 with suitable ρ and δ. Their efficiency is tunable: with appropriate ρ, ARE can exceed 90% under normality **?**. However, computation can be intensive, with cost depending on the implementation and tuning procedure.

High-breakdown estimators often come with practical hurdles: they can be sensitive to tuning choices, and convergence may slow considerably as the sample size or dimensionality increases. Ranked Set Sampling may alter which candidate observations enter subsequent calculations, but the present study does not establish a reduction in optimization or projection cost.

2.3 Minimum Volume Ellipsoid (MVE)

The MVE estimator **?** finds the smallest ellipsoid (minimum determinant of $\boldsymbol{\Sigma}$) containing at least h data points:

$$\#\left\{i : (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \le c\right\} \ge h$$

with c chosen for consistency. Asymptotically, the breakdown point approaches 0.5. However, MVE has low efficiency (typically $< 50\%$ at normality) and is computationally more demanding than MCD ($O(n^3)$ for naive search).

Despite their strong robustness, high-breakdown estimators can be computationally demanding, particularly as sample size and dimension increase. RSS changes which observations are measured, but the present study does not establish a reduction in the optimization or fitting cost of these estimators.

2.4 Stahel-Donoho Estimator

The Stahel-Donoho estimator **?** quantifies outlyingness as

$$o(\mathbf{x}_i) = \sup_{\|\mathbf{u}\|=1} \frac{|\mathbf{u}^\top \mathbf{x}_i - \text{med}(\mathbf{u}^\top X)|}{\text{MAD}(\mathbf{u}^\top X)},$$

using the direction $\mathbf{u}$ where the standardized deviation is maximized. In practice, random projections approximate the supremum, yielding $O(n^2 p)$ complexity. Breakdown point approaches 0.5 with enough projections. Efficiency is moderate and can be improved with reweighting.

Projection-based estimators often depend heavily on both the number and the choice of projection directions, and their computational cost can quickly escalate as n and p grow. RSS may alter which observations enter a subsequent estimator, but any reduction in projection or optimization cost is an empirical question not addressed by the present study.

Table 1: Comparison of High-Breakdown Estimators

Estimator	**Breakdown Point**	**Efficiency (Normal)**	**Computational Cost**	**RSS Integration Potential**
MCD	≈ 0.5	Moderate ($\sim$85%)	$O(n^2 p)$ (FAST-MCD)	Subset selection
S-Estimator	≈ 0.5	High ($> 90\%$) (tunable)	$O(n^2 p^2)$ per iteration	Subset/projection selection
MVE	≈ 0.5	Low ($< 50\%$)	$O(n^3)$	Subset selection
Stahel-Donoho	≈ 0.5	Moderate	$O(n^2 p)$ (random proj.)	Projection/subset selection

In summary, high-breakdown estimators play a critical role in ensuring robustness against data contamination, each offering unique strengths and trade-offs. Their combination with efficient data collection methods—as detailed in subsequent sections—forms a promising strategy for enhancing both robustness and statistical performance in practical applications.

3 Ranked Set Sampling under Imperfect Ranking

Ranked Set Sampling (RSS) exploits inexpensive ranking information to select a measured sample that is deliberately spread across rank strata. Its statistical benefit depends on the quality of that ranking. In the present framework, ranking quality is represented by a transition mechanism between true and observed ranks rather than by a single correlation coefficient.

3.1 Balanced RSS and an imperfect-ranking model

Consider a continuous scalar variable X with distribution function F, mean μ, and variance $\sigma^2 < \infty$. In a balanced RSS cycle, m independent sets of size m are ranked and one unit is

selected from each observed rank $s \in \{1, \ldots, m\}$. After k independent cycles, the measured sample size is $n = km$.

Let $R \in \{1, \ldots, m\}$ denote the true rank of a unit within its set and let $\widetilde{R}$ denote its observed rank. We describe imperfect ranking by the transition matrix

$$P_{rs} = \Pr(\widetilde{R} = s \mid R = r), \qquad r, s = 1, \ldots, m. \tag{1}$$

Each row of P sums to one. The matrix contains substantially more information than a scalar rank correlation: it records which true strata are confused with which observed strata.

For a fixed observed rank s, let

$$q_{r|s} = \Pr(R = r \mid \widetilde{R} = s) = \frac{P_{rs}}{\sum_{u=1}^{m} P_{us}}, \tag{2}$$

whenever the denominator is positive. If $X_{(r)}$ denotes the rth order statistic from a set of size m, the selected observation at observed rank s has density

$$g_s(x) = \sum_{r=1}^{m} q_{r|s} f_{(r)}(x), \tag{3}$$

where $f_{(r)}$ is the density of $X_{(r)}$. Equation (3) makes explicit how ranking errors change the distribution actually sampled.

Proposition 3.1 (Balanced RSS under a doubly stochastic ranking mechanism)**.** *Suppose P is doubly stochastic, so that both its row and column sums equal one. Then balanced RSS has the following properties.*

1. *The average selected distribution is the population distribution:*

$$\frac{1}{m} \sum_{s=1}^{m} g_s(x) = f(x).$$

   Consequently, the balanced RSS sample mean is unbiased for μ.

2. *If the cycles are independent, the exact variance of the RSS sample mean is*

$$\mathrm{Var}(\bar{X}_{RSS}) = \frac{1}{km^2} \sum_{s=1}^{m} \mathrm{Var}_{g_s}(X). \tag{4}$$

3. *If $P_{rs} = 1/m$ for all r, s, then $g_s = f$ for every s and balanced RSS reduces, in distribution, to SRS of size n.*

4. *If $P = I_m$, then $g_s = f_{(s)}$ and the usual perfectly ranked balanced RSS design is recovered.*

Proof. For a doubly stochastic matrix, $\sum_{s=1}^{m} P_{rs} = \sum_{s=1}^{m} q_{r|s} = 1$ because each column sum is one. Hence

$$\frac{1}{m} \sum_{s=1}^{m} g_s(x) = \frac{1}{m} \sum_{r=1}^{m} f_{(r)}(x) = f(x),$$

where the last equality is the standard order-statistic identity. The variance expression follows by independence of the km selected observations grouped by observed rank. Uniform P gives $g_s = f$, while $P = I_m$ gives $g_s = f_{(s)}$. □

The doubly stochastic condition is sufficient, not necessary, for unbiasedness. It is imposed here because it gives a transparent design-level condition and includes the two important limiting cases: perfect ranking and completely random ranking. More general ranking mechanisms can be handled directly through the weights in Eq. (2), but unbiasedness and efficiency must then be checked for the particular mechanism.

3.2 Efficiency and ranking quality

For the balanced design, define

$$\mathrm{RE}(P) = \frac{\mathrm{Var}(\bar{X}_{SRS})}{\mathrm{Var}(\bar{X}_{RSS})} = \frac{m\sigma^2}{\sum_{s=1}^{m} \mathrm{Var}_{g_s}(X)}, \tag{5}$$

where $\mathrm{Var}(\bar{X}_{SRS}) = \sigma^2/(km)$. This is an exact finite-sample expression under Proposition 3.1. It shows directly why a scalar correlation cannot, in general, determine RSS efficiency: the right-hand side depends on the entire collection of selected-stratum distributions $\{g_s\}$.

A scalar rank-correlation measure remains useful for reporting ranking quality. We use

$$\rho_R = \mathrm{Corr}(R, \widetilde{R}), \tag{6}$$

as a summary statistic, not as a sufficient parameter for efficiency. In particular, the same value of ρ_R can correspond to different transition matrices and therefore different values of $\mathrm{RE}(P)$.

Remark 3.1 (Limiting cases)**.** *Perfect ranking corresponds to $P = I_m$ and gives the classical balanced RSS variance. Random ranking corresponds to $P_{rs} = 1/m$ and gives $\mathrm{RE}(P) = 1$. Intermediate mechanisms need not produce an efficiency curve determined uniquely by ρ_R. An empirical relationship between ρ_R and efficiency is therefore a property of a specified ranking-error model, not a distribution-free theorem.*

3.3 Multivariate RSS

Let $\mathbf{X} \in \mathbb{R}^p$. A multivariate RSS design requires a scalar ranking score $R(\mathbf{x})$, such as a marginal score, a robust distance, or a projection-based outlyingness measure. The ranking score induces a partition of each set into ordered strata, but the resulting strata need not correspond to a natural order for every component of $\mathbf{X}$.

For a scalar target $a^\top \mathbf{X}$, the appropriate efficiency quantity is

$$\mathrm{RE}_a(P) = \frac{\mathrm{Var}(a^\top \bar{\mathbf{X}}_{SRS})}{\mathrm{Var}(a^\top \bar{\mathbf{X}}_{RSS})}. \tag{7}$$

There is no general guarantee that $\mathrm{RE}_a(P) \geq 1$ for every direction a, ranking score, dimension p, or contamination configuration. This is precisely the regime that the simulations investigate.

A robust multivariate ranking should be specified independently of the final estimator. Examples include robust Mahalanobis distance, projection-based outlyingness, and depth scores. For a robust Mahalanobis score, the centered form

$$R(\mathbf{x}) = (\mathbf{x} - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{x} - \hat{\boldsymbol{\mu}})$$

should be used. Its usefulness depends on how reliably the pilot estimates $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ can be obtained in the relevant dimension and contamination regime.

3.4 Robust estimation after RSS selection

Let T be a robust estimator computed from the measured RSS observations. The key distinction is between (i) robustness of T conditional on the selected sample and (ii) robustness of the complete selection-plus-estimation procedure. A robust estimator cannot by itself guarantee the latter if the ranking mechanism systematically directs contaminated observations into the measured sample.

We therefore define the ranking mechanism to be *selection-robust at contamination level* ϵ if, under the specified contamination model, the distribution of selected observations remains in a class for which the breakdown or bias bound of T is valid. This is a model-dependent condition and must not be replaced by the statement that the ranking function merely has a high breakdown point.

Proposition 3.2 (Conditional robustness transfer)**.** *Let T_n have finite-sample breakdown point $\epsilon_n^*(T)$ on a class $\mathcal{F}$ of samples. Suppose the complete RSS selection mechanism, including its ranking rule, maps every contaminated sample with at most $\lfloor n\epsilon \rfloor$ replacements into a selected sample belonging to the admissible class for T_n whenever $\epsilon < \epsilon_n^*(T)$. Then the RSS estimator T_{RSS} does not break down on that class for contamination levels below $\epsilon_n^*(T)$.*

Proof. Under the stated selection condition, the selected sample remains within the class on which T_n is bounded. The definition of finite-sample breakdown therefore implies that T_{RSS} remains bounded for the specified contamination levels. No stronger unconditional breakdown claim follows without an explicit condition guaranteeing the stated selection property. □

Proposition 3.2 is intentionally conditional. It separates a property of the base robust estimator from the additional requirement imposed by the ranking stage. In particular, statements such as "RSS preserves the breakdown point" are justified only after the selection-robustness condition has been established for the ranking mechanism and contamination model under consideration.

3.5 Influence-function perspective

For a robust estimator admitting an asymptotic linear representation under an appropriate sampling design,

$$\sqrt{n}\{T_n - \theta\} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathrm{IF}(X_i; T, F) + o_p(1).$$

Under RSS, the summands are not generally identically distributed because their distributions depend on the selected rank stratum. Consequently, the asymptotic variance is determined by the selected-stratum laws, not simply by replacing F with an order-statistic density in an i.i.d. influence-function formula. For balanced RSS with independent cycles, a

design-specific variance expression can be obtained from the covariance of the influence-function contributions in each selected stratum. A universal inequality comparing this variance with the SRS variance requires additional assumptions and is not asserted here.

This distinction is important for nonlinear estimators such as MCD, S-estimators, MVE, and Stahel–Donoho procedures. Their finite-sample behavior depends on the geometry of the selected sample and on the particular implementation. Accordingly, the present framework treats efficiency gains for these estimators as empirical quantities unless a separate estimator-specific theorem is available.

3.6 Adaptive RSS–SRS decision rule

The preceding results suggest an operational principle: RSS should be used when ranking information is sufficiently informative and robust, and SRS should be retained otherwise. Let $\widehat{V}_{RSS}$ and $\widehat{V}_{SRS}$ be pilot estimates of the variance (or MSE, if contamination is part of the target criterion) of the final estimator under the two designs. A simple decision rule is

$$D^* = \begin{cases} \text{RSS}, & \widehat{V}_{RSS} + \lambda \widehat{C}_{RSS} < \widehat{V}_{SRS} + \lambda \widehat{C}_{SRS}, \\ \text{SRS}, & \text{otherwise}, \end{cases} \tag{8}$$

where $\widehat{C}$ measures computational or measurement cost and $\lambda \geq 0$ specifies the relative cost weight. When only statistical precision matters, set $\lambda = 0$. A pilot can also report ρ_R and the estimated transition matrix $\widehat{P}$; these quantities are diagnostics, whereas Eq. (8) is the final decision criterion.

The adaptive formulation changes the scientific question from "How much does RSS improve efficiency?" to "Under which ranking and contamination regimes does RSS improve efficiency enough to justify its use?" This question is directly testable and accommodates the possibility of negative or negligible RSS gains.

4 Methodology and Implementation

This section specifies the implementation used to evaluate the framework. The experimental unit is a complete RSS cycle containing m sets and m measured observations in balanced designs. The same total number of measured observations is used for RSS and SRS comparisons. Ranking cost is reported separately from estimation cost so that statistical efficiency is not confused with computational speed.

4.1 Ranking mechanisms

The primary V3 experiment uses two explicit imperfect-ranking mechanisms. For a set with true rank vector $R = (R_1, \ldots, R_m)$, the observed ordering is generated from perturbed rank scores

$$S_i = R_i + \tau Z_i,$$

where $Z_i \sim N(0,1)$ for the Gaussian mechanism and $Z_i \sim \text{Laplace}(0,1)$ for the Laplace mechanism. The scale τ is calibrated separately for each target ranking quality so that the

expected rank correlation is approximately

$$\rho_R \in \{0.25, 0.50, 0.75, 0.90\}.$$

Calibration uses 20,000 independent trials for each mechanism–target pair. The realized rank correlation is always computed from the generated ranks and is not treated as exactly equal to the target value.

The contamination experiments distinguish two scientifically different placements. Under *post-ranking* contamination, ranking is performed on the uncontaminated responses and contamination is introduced afterward, so the ranking mechanism itself is unchanged. Under *pre-ranking* contamination, the contaminated responses are present during ranking, so contamination can alter both the selection mechanism and the measured sample. The primary V3 experiment uses $\epsilon \in \{0, 0.10\}$ and an outlier magnitude of eight standard normal units.

4.2 Robust ranking and estimators

The primary adaptive V3 experiment is univariate and therefore does not evaluate multivariate robust Mahalanobis ranking rules. It compares the sample mean, a 10% trimmed mean, and a Huber M-estimator with tuning constant $c_H = 1.345$. These choices isolate the interaction between ranking quality, contamination, and estimator robustness within the final protocol.

The same estimator definition is used for the RSS and SRS samples within each condition. No information from the independent fixed-strategy reference experiment is available to the adaptive decision; the reference experiment is used only after the adaptive run to quantify oracle agreement and fixed-strategy risk.

4.3 Adaptive selection

Algorithm 1 summarizes the proposed adaptive procedure.

Algorithm 1 Adaptive RSS–SRS Robust Estimation

1: **Input:** set size m, estimator T, ranking rule R, pilot budget, and cost weight λ
2: Draw a pilot sample and obtain ranking diagnostics $(\widehat{P}, \widehat{\rho}_R)$
3: Estimate the precision (or MSE) and cost of RSS and SRS on the pilot
4: **if** $\widehat{V}_{RSS} + \lambda \widehat{C}_{RSS} < \widehat{V}_{SRS} + \lambda \widehat{C}_{SRS}$ **then**
5: Use balanced RSS for the measurement stage
6: **else**
7: Use SRS for the measurement stage
8: **end if**
9: Apply T to the resulting measured sample
10: **Return:** estimate, ranking diagnostics, and design decision

For the primary V3 experiment, the decision criterion is the pilot variance alone, i.e. $\lambda = 0$, and the pilot does not use the known simulation truth to estimate MSE. Each adaptive trial

first generates an independent pilot with 200 pilot replications and 20 RSS cycles per pilot replication, chooses RSS or SRS from the two pilot variance estimates, and then evaluates that decision on an independent dataset. The final adaptive evaluation uses 1,000 such decision trials for each condition. An independent fixed-strategy reference experiment with 5,000 replications per condition is used only ex post to define the variance and MSE oracle designs.

The rule is not claimed to be uniformly optimal. Its role in the current study is to quantify how often a finite pilot recovers the better fixed design and how much performance is lost when the pilot selects the other design. A cost-aware rule and an MSE-aware adaptive rule remain future extensions.

4.4 Evaluation criteria

For a scalar parameter, relative variance efficiency is

$$\mathrm{RE}_V = \frac{\mathrm{Var}(\widehat{\theta}_{SRS})}{\mathrm{Var}(\widehat{\theta}_{RSS})}.$$

Under contamination, we additionally report relative MSE,

$$\mathrm{RE}_M = \frac{\mathrm{MSE}(\widehat{\theta}_{SRS})}{\mathrm{MSE}(\widehat{\theta}_{RSS})},$$

because variance and MSE need not agree when bias is present. Values above one favor RSS, values below one favor SRS, and values near one indicate little practical difference.

For each Monte Carlo configuration, the study reports fixed-strategy variance and MSE references together with the adaptive selected-design variance and MSE. In the primary V3 adaptive rule, the pilot minimizes sampling variance; MSE is evaluated only ex post. The present results therefore do not establish an MSE-optimal adaptive rule under contamination. Monte Carlo uncertainty for the final adaptive and fixed-reference metrics is reported in Section 5; condition-level intervals are provided in the accompanying analysis output. Breakdown is not inferred from the finite grid of contamination percentages; a breakdown claim requires a separate contamination experiment that searches for divergence or instability under the stated contamination model.

4.5 Computational accounting

The total cost is conceptually decomposed as

$$C_{\text{total}} = C_{\text{ranking}} + C_{\text{sorting}} + C_{\text{estimation}}.$$

The V3 adaptive experiment does not report wall-clock timing, so no computational-speed conclusion is drawn from the present results. A computational study would need to benchmark per-estimate cost for RSS and SRS under identical hardware and software, and should distinguish ranking/sorting cost from estimator cost and from any precision-matched Monte Carlo savings.

4.6 Reproducibility and audit trail

The final primary results are tied to a fixed computational protocol and are reported from the final data products rather than manually transcribed exploratory analyses. The adaptive experiment uses seed 20260915, 96 conditions, 200 pilot replications, 20 pilot cycles, and 1,000 adaptive evaluation trials per condition. The independent fixed-strategy reference experiment uses 5,000 replications per condition. The uncertainty analysis uses 10,000 nonparametric bootstrap replicates with bootstrap seed 20260915 and Wilson intervals for trial-level proportions. The reproducibility package contains the exact simulation drivers, the final adaptive trial-level CSV, the final 5,000-reference CSV, the rebuilt 96-row summary, the uncertainty-analysis outputs, and a README mapping each primary table and reported scalar to its source file and analysis step. Software-package versions and the execution environment should be archived with the submission package rather than inferred from the manuscript.

5 Simulation Study

The primary numerical evidence is a calibrated adaptive V3 experiment designed to test the proposed RSS–SRS decision rule under controlled ranking quality and contamination placement. The experiment is deliberately narrower than the full theoretical framework: it is univariate, uses $m = 5$, and compares the mean, 10% trimmed mean, and Huber M-estimator. The numerical evidence reported below is restricted to the final V3 protocol; earlier multivariate and benchmark-data analyses are excluded from the primary evidence chain because they were generated under different protocols.

5.1 Primary V3 adaptive experiment

Each condition uses $m = 5$ sets per cycle, 100 cycles, and hence $n = 500$ measured observations. The ranking targets are $\rho_R \in \{0.25, 0.50, 0.75, 0.90\}$; the ranking-error mechanism is Gaussian or Laplace; contamination is absent or has fraction $\epsilon = 0.10$; and contamination is placed either before or after ranking. The three estimators are the sample mean, 10% trimmed mean, and Huber M-estimator. This produces $4 \times 2 \times 2 \times 2 \times 3 = 96$ adaptive conditions.

The ranking scales are calibrated using 20,000 trials per target–mechanism pair. Table 2 shows that the achieved calibration targets are accurate to numerical precision at $m = 5$.

For every condition, the final adaptive pilot uses 200 independent pilot replications with 20 cycles per replication. The final adaptive evaluation contains 1,000 independent pilot–decision–evaluation trials. The fixed-strategy reference risks are estimated independently from 5,000 replications per condition. Importantly, the adaptive decision uses only the pilot variance estimates; the fixed-strategy reference risks are used only for ex-post diagnostics.

Table 2: V3 ranking-scale calibration at $m = 5$ (20,000 calibration trials per row).

Mechanism	Target ρ_R	Calibrated scale τ	Pilot expected ρ_R	Absolute error
Gaussian	0.25	5.56036	0.24999	1.0×10^{-5}
Laplace	0.25	4.76666	0.25000	$< 10^{-12}$
Gaussian	0.50	2.50672	0.50000	$< 10^{-12}$
Laplace	0.50	2.07187	0.50000	$< 10^{-12}$
Gaussian	0.75	1.33957	0.75000	$< 10^{-12}$
Laplace	0.75	1.03538	0.75000	$< 10^{-12}$
Gaussian	0.90	0.795978	0.90000	$< 10^{-12}$
Laplace	0.90	0.579369	0.900005	5.0×10^{-6}

5.1.1 Fixed-strategy reference risks

The independent reference experiment provides a stable benchmark against which the adaptive decisions can be interpreted. Define

$$\mathrm{RE}_{V,\mathrm{ref}} = \frac{V_{SRS}}{V_{RSS}}, \qquad \mathrm{RE}_{M,\mathrm{ref}} = \frac{MSE_{SRS}}{MSE_{RSS}}.$$

Values above one indicate that RSS has lower fixed-strategy risk, while values below one indicate the opposite. Averaged over the 24 conditions at each target ρ_R, the mean reference variance-efficiency ratios increase from 1.028 at $\rho_R = 0.25$ to 1.170, 1.477, and 1.884 at $\rho_R = 0.50, 0.75, 0.90$, respectively. The reference efficiency remains heterogeneous across mechanism, contamination placement, and estimator. Thus, even in this controlled study, ranking quality does not induce a universal efficiency guarantee. Across all 96 conditions, the mean reference variance-efficiency ratio is 1.3897, the median is 1.2253, and the range is 0.9622–2.1858.

Table 3: Independent V3 fixed-strategy reference risks summarized by target ranking quality (5,000 reference replications per condition).

Target ρ_R	Mean RE_V
0.25	1.028
0.50	1.170
0.75	1.477
0.90	1.884

Across all 96 conditions, the fixed-strategy oracle is RSS in 90 conditions and SRS in 6 conditions, for both variance and MSE because the same oracle method is obtained under the two reference criteria. The six SRS-oracle conditions are all in the weakest ranking regime ($\rho_R = 0.25$), demonstrating why the fallback option is necessary.

5.1.2 Adaptive selection performance

The primary adaptive outcome is the fraction of the 1,000 pilot-driven trials in which RSS is selected. The final fixed-strategy oracle is computed from 5,000 independent reference

Table 4: Sensitivity of adaptive selection to pilot budget, using 1,000 adaptive trials per condition and the 5,000-replicate fixed-strategy oracle.

Pilot reps	Overall RSS selection	Median condition selection	Trial oracle agreement	Majority oracl
10	0.6549	0.5880	0.6524	89/96 =
20	0.7051	0.6545	0.7030	89/96 =
30	0.7363	0.6935	0.7342	90/96 =
50	0.7690	0.7520	0.7664	90/96 =
100	0.8080	0.8330	0.8042	91/96 =
200	0.8397	0.9075	0.8354	90/96 =

replications per condition. Because the pilot-budget sensitivity study shows that the selection fraction changes materially with pilot size, the final manuscript reports the selected 200-replication pilot budget explicitly rather than presenting the selection fraction as pilot-budget invariant.

Table 5: Final adaptive V3 performance with 200 pilot replications and 1,000 adaptive trials per condition. Wilson 95% intervals are reported for trial-level proportions.

Metric	Final value (95% CI where applicable)
Overall RSS selection fraction	0.839729 [0.837395, 0.842036]
Median condition RSS selection fraction	0.9075
Trial-level variance-oracle agreement	0.835375 [0.833016, 0.837707]
Majority condition-level oracle agreement	90/96 = 93.75%

With the final 200-replication pilot and 1,000 adaptive trials per condition, the overall RSS selection fraction is 0.839729 (95% Wilson CI: 0.837395–0.842036), and the median condition-level selection fraction is 0.9075. Agreement with the 5,000-replicate fixed-strategy variance oracle is 0.835375 at the trial level (95% Wilson CI: 0.833016–0.837707), while the majority adaptive choice agrees with the oracle in 90 of 96 conditions (93.75%). These quantities describe the finite-pilot decision behavior for the specified 200-replication pilot and should not be interpreted as an MSE-optimality result.

The final 5,000-replicate fixed-strategy reference experiment gives a mean variance relative efficiency of 1.389747 (bootstrap 95% CI: 1.382332–1.398196) and a median of 1.225325 (bootstrap 95% CI: 1.213007–1.258669). The corresponding MSE relative-efficiency point estimates are 1.389689 for the mean and 1.225473 for the median. The fixed-reference variance oracle selects RSS in 90 of 96 conditions and SRS in 6 of 96 conditions. These intervals quantify Monte Carlo uncertainty in the reference experiment; they do not convert the ex-post reference oracle into a decision available to the adaptive procedure.

Negative values of adaptive excess risk in individual rows are not interpreted as evidence that the adaptive procedure systematically beats the oracle. The adaptive risks use 1,000 evaluation trials, whereas the oracle references use 5,000 fixed-strategy replications; sampling variability can therefore make an adaptive estimate fall below the reference estimate. Final submission tables should attach uncertainty intervals before making row-level comparisons.

5.1.3 Matched-mechanism comparison

A separate paired V3 experiment was used to test the central theoretical claim that a scalar rank correlation does not uniquely determine efficiency. Gaussian and Laplace ranking mechanisms were simulated at closely matched realized rank correlations, using the same paired random seeds. Across $m = 5$, 100 cycles, 500 measured observations, and four estimators (mean, trimmed, Huber, and Tukey), the realized correlations differ by at most 0.0039 at the four target levels. The mean variance-efficiency difference (Gaussian minus Laplace) is -0.052 at $\rho_R = 0.25$, +0.151 at 0.50, +0.143 at 0.75, and -0.053 at 0.90.

The bootstrap intervals show that the evidence for a mechanism effect is strongest around $\rho_R = 0.50$: one of four estimator-specific variance contrasts has a 95% bootstrap interval excluding zero, and three of four MSE contrasts exclude zero. At $\rho_R = 0.25$, 0.75, and 0.90, the estimator-specific intervals all cross zero. The result therefore supports mechanism dependence as a measurable phenomenon without supporting a universal direction or magnitude of the effect.

Table 6: Paired Gaussian-versus-Laplace comparison at matched ranking quality. Differences are Gaussian minus Laplace in relative efficiency.

Target ρ_R	Matched realized correlations	Mean ΔRE_V	Range of ΔRE_V	Variance CIs excluding 0
0.25	0.24696 / 0.24929	-0.052	[-0.057,-0.043]	0/4
0.50	0.50167 / 0.49930	+0.151	[0.146,0.159]	1/4
0.75	0.75266 / 0.74880	+0.143	[0.139,0.151]	0/4
0.90	0.90041 / 0.90096	-0.053	[-0.070,-0.039]	0/4

The V3 evidence therefore supports the paper's main design-level point in a qualified form: the same approximate scalar ranking correlation can coexist with different empirical efficiency, but the current experiment is not sufficient to map the full transition-matrix space or to establish a universal ordering among mechanisms.

5.1.4 Scope of the current numerical evidence

The V3 experiment is now the primary confirmatory simulation in the manuscript. The adaptive-trial budget was increased from 200 to 1,000, and the first 200 trials of the 1,000-trial run were verified to be exactly identical to the preserved 200-trial run. At the aggregate level, increasing the adaptive-trial budget changed the overall RSS-selection fraction from 0.7414 to 0.7363 (0.51 percentage points) and trial-level oracle agreement from 0.7377 to 0.7342 (0.35 percentage points); the majority condition-level oracle agreement increased from 91.67% to 94.79%, with 7 of 96 conditions changing majority choice.

Pilot-budget sensitivity was also evaluated at 10, 20, 30, 50, 100, and 200 pilot replications, using 1,000 adaptive trials per condition. The overall RSS-selection fraction increased from 0.6549 at 10 pilots to 0.8397 at 200 pilots, while trial-level oracle agreement increased from 0.6524 to 0.8354. Thus, the pilot budget is a material tuning parameter rather than a negligible implementation detail; the final primary experiment uses 200 pilot replications because it is the largest tested budget, but the study does not claim pilot-budget invariance.

Perfect-ranking and random-ranking controls, multiple master seeds, a cost-aware decision rule, and an MSE-aware pilot criterion remain outside the present V3 validation. They are not used to support the primary confirmatory claims.

6 Conclusion

This paper develops a design-based framework for Ranked Set Sampling under imperfect ranking and a corresponding adaptive RSS–SRS selection rule. The ranking mechanism is represented explicitly by a transition matrix rather than compressed into a single scalar rank-correlation summary. Under the stated balanced-RSS assumptions and the doubly stochastic condition used in the analysis, the paper derives exact finite-sample expressions for the mean and variance of the RSS mean. These results clarify which conclusions follow from the sampling design and which require additional assumptions. The robustness discussion is deliberately conditional: robustness of a base estimator is not asserted to transfer automatically through the ranking and selection mechanism.

The primary numerical evidence is the final V3 experiment. It contains 96 univariate conditions with $m = 5$ and 100 cycles, corresponding to $n = 500$ measured observations per trial. Conditions vary over four target ranking qualities, Gaussian or Laplace ranking-error mechanisms, contamination levels
$epsilon \in \{0, 0.10\}$, contamination placement before or after ranking, and three estimators. The adaptive procedure uses an independent 200-replication pilot and 1,000 adaptive evaluation trials per condition; fixed-strategy reference risks are estimated independently from 5,000 replications per condition. Under this specified protocol, RSS was selected in 0.839729 of adaptive trials (95% Wilson CI $[0.837395, 0.842036]$), and the selected method agreed with the 5,000-replicate variance oracle in 0.835375 of trials (95% Wilson CI $[0.833016, 0.837707]$). At the condition level, the majority adaptive choice matched the variance oracle in 90 of 96 conditions. The reference variance-efficiency ratio had mean 1.389747 (bootstrap 95% CI $[1.382332, 1.398196]$) and median 1.225325 (bootstrap 95% CI $[1.213007, 1.258669]$).

The matched-mechanism experiment provides complementary evidence that approximately equal scalar rank correlations can coexist with different empirical efficiencies. The observed differences between Gaussian and Laplace ranking mechanisms varied in sign and magnitude across the tested ranking-quality levels. This supports the narrower claim that a scalar rank-correlation summary does not uniquely determine empirical efficiency within the mechanisms examined; it does not establish a universal ordering of ranking mechanisms.

The adaptive results must be interpreted as finite-sample characteristics of the specified protocol. Pilot-budget sensitivity shows that selection behavior changes materially as the pilot size varies, so 200 pilot replications are a protocol choice rather than a universally optimal budget. The adaptive criterion is variance-based and is not an MSE-optimality theorem. The 5,000-replicate reference oracle is an ex-post benchmark and is not available to the adaptive rule during selection. Perfect-ranking and random-ranking controls and independent master-seed replications were not included in the final V3 run, so broader claims of robustness to ranking extremes or random-seed variation are outside the present evidence.

The theoretical scope is similarly explicit. The manuscript establishes a sufficient doubly stochastic condition for unbiasedness and an exact variance formula within the stated model; it does not claim a necessary-and-sufficient characterization for arbitrary transition matrices, a general exact variance formula outside the stated setting, a finite-pilot selection-error theorem, an adaptive regret bound, or a general estimator-specific robustness theorem. Establishing such results would require additional analysis beyond the present paper. Accordingly, the empirical adaptive rule is presented as a calibrated finite-sample procedure rather than as a theoretically optimal decision rule.

Earlier multivariate and benchmark-data analyses are excluded from the primary evidence chain because they were generated under earlier protocols. They should not be interpreted as validation of the final V3 adaptive claims. The reproducibility package records the exact final simulation drivers, raw adaptive trials, 5,000-replicate reference data, uncertainty-analysis scripts, seed information, and a table-to-source mapping so that the reported primary results can be regenerated and audited independently.

Overall, the evidence supports a conditional conclusion: under the specified V3 design, ranking quality and ranking mechanism materially affect whether RSS improves fixed-strategy precision, and an independently piloted RSS–SRS rule can adapt its sampling choice to the estimated variance. The results do not establish universal superiority of RSS over SRS, optimality of the chosen pilot size, MSE optimality of the adaptive rule, computational superiority, or stability across arbitrary master seeds.

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Competing Interests

The authors declare that they have no competing interests.